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Course Code: CSA1718

PROGRAM – 5

Missionaries and Cannibals Problem

Aim

To solve the Missionaries and Cannibals problem using the Breadth First Search (BFS) algorithm.

Algorithm

1. Represent each state as (Missionaries Left, Cannibals Left, Boat Position).
2. Start from the initial state (3, 3, Left).
3. Generate all valid successor states.
4. Check whether missionaries are never outnumbered by cannibals on either river bank.
5. Use Breadth First Search (BFS) to explore valid states.
6. Stop when the goal state (0, 0, Right) is reached.
7. Display the solution path.

Python Program

```
from collections import deque
m=[(2,0),(0,2),(1,1),(1,0),(0,1)]
v=lambda a,b:0<=a<=3 and 0<=b<=3 and(a==0 or a>=b)and((3-a)==0 or(3-a)>=(3-b))
q=deque([((3,3,1),[(3,3,1)]))
s={(3,3,1)}
while q:
    (a,b,c),p=q.popleft()
    if(a,b,c)==(0,0,0):
        print(p)
        break
    for x,y in m:
        n=(a-x,b-y,0)if c else(a+x,b+y,1)
        if v(n[0],n[1])and n not in s:
            s.add(n)
            q.append((n,p+[n]))
```

Expected Output

Solution Path:

(3, 3, 1)

(3, 1, 0)

(3, 2, 1)

(3, 0, 0)

(3, 1, 1)

(1, 1, 0)

(2, 2, 1)

(0, 2, 0)

(0, 3, 1)

(0, 1, 0)

(1, 1, 1)

(0, 0, 0)

PROGRAM – 6

Vacuum Cleaner Problem

Aim

To simulate the Vacuum Cleaner Agent using Artificial Intelligence.

Algorithm

1. Initialize two rooms A and B.
2. Check the current room.
3. If the room is dirty, clean it.
4. Move to the next room.
5. Repeat until both rooms are clean.
6. Display the final status.

Python Program

```
rooms={'A':'Dirty','B':'Dirty'}
loc='A'

while 'Dirty' in rooms.values():
    if rooms[loc]=='Dirty':
        print("Cleaning",loc)
        rooms[loc]='Clean'
    loc='B' if loc=='A' else 'A'

print("Final State:")
print(rooms)
```

Expected Output

```
Cleaning A
Cleaning B
Final State:
{'A': 'Clean', 'B': 'Clean'}
```